

STRAUBING, Germany

WMO: 107880

Lat: 48.8275N

Lon: 12.5597E

Elev: 359

StdP: 97.08

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-10.8	-16.1	1.0	-13.6	-12.9	1.3	-10.3	11.4	7.4	10.0	6.3	1.4	90	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.6	19.6	27.7	18.9	26.2	18.4	20.7	27.3	19.9	25.9	19.1	24.6	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.0	23.5	17.7	13.3	22.3	17.0	12.7	21.4	61.5	27.6	58.4	26.0	55.7	24.8	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.0	5.9	-17.4	32.6	3.7	1.5	-20.1	33.7	-22.2	34.6	-24.3	35.5	-26.9	36.6		
(5)			-17.7	22.8	3.6	0.9	-20.3	23.4	-22.5	23.9	-24.5	24.4	-27.1	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-1.6	0.2	4.4	9.3	13.9	17.0	18.6	18.2	13.8	9.1	3.6	-0.5		
	DBStd	8.13	4.84	4.97	4.02	3.94	3.49	3.50	2.96	2.89	3.13	3.66	3.65	4.37		
	HDD10.0	1459	359	275	177	59	7	1	0	0	4	61	194	324		
	HDD18.3	3561	617	508	432	272	142	67	34	37	139	287	442	582		
	CDD10.0	1051	0	0	3	37	128	209	266	256	118	33	2	0		
	CDD18.3	111	0	0	0	1	6	26	42	34	3	0	0	0		
	CDH23.3	1192	0	0	0	7	90	271	421	364	38	1	0	0		
	CDH26.7	279	0	0	0	1	12	59	116	91	2	0	0	0		
(14) Wind	WSAvg	2.5	2.6	2.8	3.0	2.7	2.5	2.3	2.3	2.1	2.3	2.3	2.5	2.6		
(15) Precipitation	PrecAvg	861	64	57	61	47	82	97	99	86	70	62	66	70		
	PrecMax	1160	143	120	160	104	176	149	160	160	160	182	134	159		
	PrecMin	617	7	16	11	5	12	40	12	27	30	16	0	17		
	PrecStd	111	33	28	38	25	36	26	36	37	30	36	31	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	12.2	17.8	24.2	28.3	30.7	32.0	32.4	26.5	22.1	15.0	10.6		
		MCWB	8.7	8.8	10.5	14.1	18.0	20.7	21.0	20.0	18.6	16.0	11.6	8.6		
	2%	DB	9.3	10.3	15.2	21.7	25.9	28.4	29.7	29.0	24.4	18.5	12.0	8.1		
		MCWB	7.2	7.5	9.9	13.2	17.3	19.8	19.7	19.0	17.6	14.7	10.0	6.8		
	5%	DB	6.9	8.4	13.0	19.4	23.7	26.5	27.7	27.1	22.3	16.4	10.3	6.2		
		MCWB	5.3	6.4	8.7	12.3	16.0	18.9	19.2	18.5	16.6	13.5	8.8	5.3		
	10%	DB	4.9	6.6	10.7	17.1	21.4	24.6	25.6	25.3	20.3	14.6	8.6	4.6		
		MCWB	3.7	5.0	7.8	10.9	14.9	17.9	18.5	18.0	15.7	12.5	7.5	3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.1	11.8	15.6	19.8	22.2	22.1	21.5	19.6	16.7	11.9	9.0		
		MCDB	11.3	11.4	16.1	22.0	26.1	28.6	29.8	28.8	24.6	20.1	14.2	10.4		
	2%	WB	7.1	7.8	10.3	13.8	17.8	20.7	20.6	20.2	18.3	15.2	10.2	6.9		
		MCDB	8.9	9.9	14.1	20.3	23.9	26.9	27.4	26.6	22.8	17.8	11.7	8.0		
	5%	WB	5.4	6.5	9.1	12.6	16.7	19.5	19.8	19.3	17.3	13.9	8.9	5.3		
		MCDB	6.8	8.2	12.2	18.4	22.3	25.0	25.9	25.2	21.1	16.1	10.0	6.1		
	10%	WB	3.7	5.1	8.0	11.5	15.6	18.4	19.0	18.5	16.2	12.7	7.6	3.9		
		MCDB	4.7	6.5	10.6	16.4	20.5	23.3	24.4	23.7	19.5	14.2	8.6	4.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.7	8.4	10.5	10.7	10.5	10.8	10.9	9.9	8.2	5.0	4.5		
		MCDBR	6.8	9.1	13.0	15.3	15.0	14.1	14.8	14.9	13.8	10.9	7.8	6.2		
	5% WB	MCWBR	5.4	6.6	8.1	8.0	7.6	6.8	6.5	6.6	7.5	7.2	5.8	5.1		
		MCWBR	6.7	8.7	11.6	14.0	13.4	13.1	13.0	13.1	11.8	9.8	7.3	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.324	0.362	0.395	0.402	0.405	0.413	0.407	0.380	0.353	0.327	0.299		
	taud		2.411	2.364	2.319	2.260	2.274	2.298	2.271	2.309	2.356	2.422	2.454	2.443		
	Ebn at Noon		761	824	846	852	859	857	843	831	817	779	724	716		
	Edh at Noon		65	86	106	125	128	126	128	118	102	81	63	56		
(44) All-Sky Solar Radiation	RadAvg		0.96	1.88	2.93	4.29	5.00	5.51	5.30	4.66	3.32	2.05	1.02	0.73		
	RadStd		0.17	0.26	0.37	0.55	0.59	0.48	0.51	0.48	0.44	0.23	0.12	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+34
(47) Regional (78 neighbors)		+0.42	N/A	N/A	N/A	+0.33	+0.37	N/A	N/A	+60	+23

Nomenclature: See separate page